Automated Quality Assessment and Adaptive Mentorship for Undergraduate Research Proposals

***Abstract*—Reviewing undergraduate research proposals requires supervisors to assess structural completeness, research reasoning, methodological adequacy, and the quality of successive revisions while providing feedback that students can act on. Repeating this process across many proposals can be time-consuming, yet fully automated grading is unsuitable because proposal quality is contextual and model outputs can be unreliable. This paper presents a human-in-the-loop framework for automated quality assessment and adaptive mentorship of undergraduate research proposals. The framework combines rule-based structural validation, semantic concept analysis, TF–IDF with a Linear Support Vector Machine (SVM) for weakness classification, SentenceTransformer embeddings for semantic retrieval of historical supervisor feedback, TF–IDF with a tuned Random Forest regressor for a baseline proposal-quality score, targeted learning-resource recommendation, and version-aware improvement tracking. The processed data contain 2,035 weakness annotations, 2,134 historical feedback records, and 165 scored grading records. The grading model obtained a mean absolute error of 5.56, root mean squared error of 6.50, and $R^2 = 0.146$. These results show that the current predictive components provide useful but limited signals and should not determine academic outcomes independently. Accordingly, the framework treats model predictions and retrieved feedback as evidence for a supervisor-controlled review workflow, with generated review drafts remaining editable before release to students. The resulting prototype demonstrates how lightweight NLP and classical machine learning can be integrated with academic oversight to support proposal review and iterative mentorship without replacing supervisor judgement.**

Index Terms*—*automated assessment, adaptive mentorship, human-in-the-loop, research proposals, semantic retrieval

I. Introduction

Undergraduate research proposals are important academic artifacts because they translate an initial research idea into a defensible problem statement, objectives, methodological plan, evaluation strategy, and expected contribution. Their quality is rarely established in one submission. Students normally improve proposals through repeated cycles of supervisor comments, revision, and re-evaluation. Effective feedback can reduce the gap between a learner's current and desired performance [1], while formative assessment can support self-regulated learning when it helps students understand how to improve rather than merely assigning a score [2].

This process creates a practical workload problem. A supervisor reviewing many proposals must repeatedly check missing sections, weak research reasoning, unclear methodology, incomplete evaluation plans, and whether earlier feedback has been addressed. Conventional submission systems preserve files and grades, but do not usually perform proposal-specific diagnostic analysis. Grammar and similarity tools address narrower concerns and cannot by themselves determine whether a proposal explains its research concepts adequately.

Natural language processing (NLP) offers useful support for parts of this workflow. Contextual language models such as BERT improved semantic text representation [3], while Sentence-BERT (SBERT) made semantic similarity search practical by producing sentence embeddings that can be compared efficiently [4]. In automated essay scoring (AES), transformer-based and classical methods have both been studied, but increased model complexity does not automatically produce more dependable scoring [5], [6]. Recent work has also emphasized the need to consider reliability, fairness, and the quality of generated feedback in automated educational assessment [7], [8].

Research-proposal assessment is especially unsuitable for an autonomous "single-score" solution. Unlike standardized essays, proposals may differ substantially by research domain, methodological paradigm, maturity, and supervisor expectations. A useful system should therefore automate repetitive analysis while exposing evidence and preserving human authority over academic judgement.

This work develops a framework titled *Automated Quality Assessment and Adaptive Mentorship for Undergraduate Research Proposals*. The system combines structural checks, semantic concept analysis, supervised classification, embedding-based feedback retrieval, baseline grading, evidence-grounded review support, adaptive learning resources, and proposal-version tracking within a human-in-the-loop workflow.

This study examines the predictive performance of the implemented weakness-classification and proposal-grading models on held-out data. It also investigates how structural analysis, semantic retrieval, model predictions, and supervisor control can be combined into a practical workflow that supports proposal review without treating automated outputs as final academic decisions.

The main contributions are: (1) a modular assessment pipeline that separates structural completeness from semantic research concepts; (2) empirical measurement of a TF–IDF/Linear-SVM weakness classifier and a TF–IDF/Random Forest grading model; (3) embedding-based retrieval of historically related supervisor feedback; (4) an explicit human-in-the-loop review stage in which model outputs serve as evidence rather than final decisions; and (5) adaptive resource recom-

mendation and version-aware tracking to connect assessment with subsequent student revision.

II. Related Work

A. Automated Writing Assessment

Automated writing assessment has progressed from surface features and hand-crafted rules to statistical and neural approaches. BERT demonstrated the value of bidirectional contextual representations for language understanding [3]. For longer student texts, Wang *et al.* showed that multi-scale representations can improve the use of BERT in AES [6]. However, Mayfield and Black found that fine-tuned BERT did not necessarily justify its additional computational cost over classical methods in AES [5]. This observation is relevant to the present work because the available task-specific datasets are comparatively small, making lightweight and inspectable models a reasonable prototype choice.

Automated scoring also introduces validity and fairness concerns. Schaller *et al.* demonstrate that average accuracy alone is insufficient when assessment models behave differently across learner groups [8]. Consequently, an educational assessment system should make its predictive limitations visible and avoid translating uncertain model outputs directly into high-stakes academic decisions.

B. Semantic Retrieval and Feedback

SBERT maps sentences into a vector space in which semantic relatedness can be estimated efficiently using cosine similarity [4]. This is useful for educational feedback because two comments may express the same criticism using different wording. In the proposed framework, embeddings are used to retrieve related historical supervisor comments, but retrieval is kept separate from validation: similarity indicates semantic proximity, not that the retrieved criticism is necessarily supported by the current proposal.

Feedback quality is central to formative learning. Hattie and Timperley argue that useful feedback should help learners understand where they are, where they need to go, and what actions can close the gap [1]. Nicol and Macfarlane-Dick similarly connect formative feedback with self-regulated learning [2]. Recent LLM-based work explores joint essay scoring and feedback generation, but also shows that generating fluent feedback is not equivalent to guaranteeing pedagogical usefulness [7]. These findings motivate the framework's combination of automated analysis, targeted learning resources, and human approval.

III. System Design and Methodology

A. Design Principles

The framework is organized around three principles. First, different kinds of proposal evidence should remain distinct. Missing headings, weak semantic content, classifier predictions, retrieved historical comments, and numerical scores have different meanings and confidence levels. Second, automated analysis should produce evidence for review rather than an irreversible academic decision. Third, assessment should connect to the student's next revision through targeted resources and version tracking.

B. Proposal Preprocessing and Versioning

Each submission is associated with a student, a proposal, and a proposal version. The implementation stores successive versions rather than overwriting a previous submission. This preserves the analysis and supervisor review linked to each iteration and provides an operational basis for comparing changes between $V_1, V_2, \ldots, V_n$. Text extraction and basic validation are performed before the content is passed to the analytical modules.

The prototype backend is implemented with FastAPI and PostgreSQL using SQLAlchemy/Alembic, while the user interface is implemented with React. The ML artifacts are serialized and loaded locally for inference. This architecture is an implementation detail rather than the research contribution; its purpose is to support a reproducible review workflow in which analyses, reviews, and proposal versions remain associated.

C. Structural Validation and Semantic Concept Analysis

Structural validation checks whether expected proposal sections can be identified, including the abstract, introduction, literature review, methodology, evaluation, and expected outcomes. It is deliberately separated from semantic concept analysis. A document may contain a heading entitled "Methodology" while failing to explain an actual methodological procedure; conversely, an evaluation concept may be discussed without appearing under an explicit "Evaluation" heading.

Semantic concept analysis therefore represents selected research concepts with three states: *present*, *implicit*, and *missing*. A concept is marked present when clear supporting text is identified, implicit when related evidence exists but articulation is incomplete, and missing when meaningful support cannot be identified. Evidence snippets can be retained for present or implicit concepts so that the interface can show why a status was assigned. Concept normalization removes obvious numeric fragments and malformed extraction artifacts before candidate concepts are surfaced to users.

D. Weakness Classification

The diagnostic classifier uses Term Frequency–Inverse Document Frequency (TF–IDF) features and a Linear SVM. For term t in document d, the feature weight follows the conventional form

$$\mathrm{tfidf}(t, d) = \mathrm{tf}(t, d) \log \frac{N}{\mathrm{df}(t)}, \tag{1}$$

where N is the number of documents and $\mathrm{df}(t)$ is the number of documents containing t. Linear SVMs are well suited to high-dimensional sparse text representations [9]. The processed weakness dataset contains 2,035 annotated records. During inference, the trained pipeline assigns weakness labels that are displayed as diagnostic signals, not definitive academic conclusions.

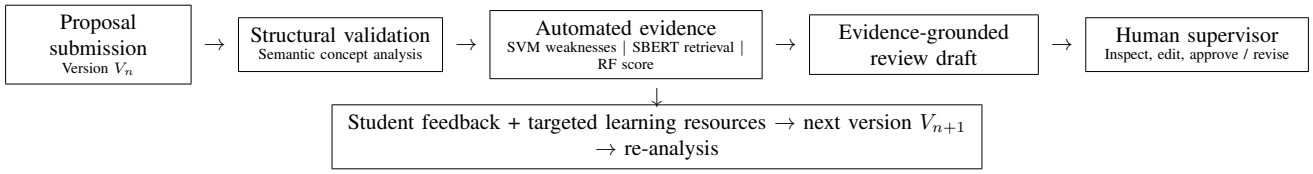


Fig. 1. Human-in-the-loop workflow for automated quality assessment and adaptive mentorship. Automated outputs enter an evidence layer; the supervisor retains control over student-facing feedback.

E. Semantic Feedback Retrieval

The retrieval component contains 2,134 historical feedback records embedded with a SentenceTransformer model. A query representation $\mathbf{q}$ is compared with a stored feedback representation $\mathbf{d}$ using cosine similarity,

$$\mathrm{sim}(\mathbf{q}, \mathbf{d}) = \frac{\mathbf{q} \cdot \mathbf{d}}{\|\mathbf{q}\|\|\mathbf{d}\|}. \tag{2}$$

The highest-ranked feedback records provide context for the review workflow. This module is intentionally described as semantic retrieval rather than an autonomous feedback generator: a similar historical comment can still include a criticism that is unsupported by the current proposal.

F. Baseline Proposal-Quality Regression

A separate component estimates a numerical proposal-quality score from 165 scored records using TF–IDF features and a tuned Random Forest regressor. Random Forests aggregate multiple decision trees and can model non-linear relationships while remaining practical on modest datasets [10]. The resulting score is presented as a baseline signal for supervisor inspection. It is not treated as the final mark because the training set is small and the measured generalization performance is limited.

G. Evidence-Grounded Review and Adaptive Mentorship

Outputs from structural validation, semantic concept analysis, weakness classification, retrieval, and grading are combined into an evidence layer. The implementation can use this evidence to construct a structured review draft covering overall assessment, methodology, evaluation, strengths, weaknesses, and recommendations. If an optional language-model reviewer is configured, its text is constrained by the same evidence and remains an editable draft. The system design therefore separates three operations that are often conflated: retrieving potentially relevant historical feedback, generating review text, and approving feedback for the student.

Adaptive mentorship is implemented through a resource-recommendation layer. Detected deficiencies are mapped to targeted learning resources. For example, a missing abstract prioritizes guidance on academic abstract writing instead of a generic methodology resource. When no recognized structural deficiency is present, broader analysis signals can be used as fallbacks. Approved review content and selected resources are delivered to the student, who can submit a new version for re-analysis.

H. Human-in-the-Loop Decision Lifecycle

The framework uses a staged decision lifecycle so that no single model output is promoted directly to student-facing feedback. In the first stage, the system collects document-grounded evidence: detected sections, concept states, and supporting snippets. In the second stage, predictive and retrieval modules add candidate interpretations, including weakness labels, related historical comments, and the baseline quality estimate. These outputs are intentionally marked as model-derived evidence because they may be incomplete or incorrect.

In the third stage, the system constructs a review draft from the available evidence. A criticism should be associated with current-proposal evidence before it is treated as a reliable recommendation. Historical comments that are semantically similar but unsupported by the current document remain contextual suggestions rather than accepted findings. The fourth stage is the human checkpoint: the supervisor can confirm, revise, delete, or add feedback and can decide whether the proposal should proceed or return for revision. Finally, approved feedback and relevant learning resources are linked to the current proposal version. A later submission creates a new version and repeats the same analysis instead of modifying the earlier evidence record.

Table I summarizes the intended responsibility boundary between automation and the supervisor. This boundary is important because the measured models are not sufficiently accurate to justify autonomous assessment; the system's design uses automation primarily to organize attention and make review evidence easier to inspect.

I. Traceability and Evidence Representation

A practical requirement of the framework is that a supervisor should be able to distinguish what was extracted directly from the proposal from what was inferred by a model. For this reason, the analysis record preserves multiple types of evidence rather than collapsing them into one score. Structural flags record whether expected sections were detected. Semantic concept records store the assigned concept state together with supporting snippets when available. Weakness predictions are stored as model outputs, while retrieved feedback is kept as a separate set of semantically related historical examples. The proposal-quality estimate is represented independently as a numerical model output.

This separation supports two forms of traceability. First, the supervisor can trace a review statement back to the current proposal evidence instead of accepting an opaque recommendation. Second, the system can preserve the origin

TABLE I
ROLE OF AUTOMATED COMPONENTS IN THE HITL WORKFLOW

Component	Automated output	Supervisor role
Structural validation	Missing/present section flags	Verify document structure
Concept analysis	Present/implicit/missing + evidence	Inspect semantic adequacy
Weakness SVM	Candidate weakness label	Confirm or reject diagnosis
Feedback retrieval	Similar historical comments	Use only when applicable
Quality regressor	Baseline numerical estimate	Treat as advisory evidence
Resource recommender	Targeted learning materials	Select/support next action

TABLE II
DATASETS AND EVALUATION STATUS

Component	Records	Held-out	Evaluation
Weakness classification	2,035	407	Accuracy, macro-F1
Feedback retrieval	2,134	–	Functional retrieval
Proposal grading	165	33	MAE, RMSE, R^2

of a suggestion, making it possible to tell whether it came from a rule, a classifier, a retrieval result, or the grading model. Such provenance becomes important when two components disagree. For example, a structurally complete methodology section may still be associated with a semantic weakness, or a retrieved historical comment may conflict with the evidence in the current proposal. The interface can therefore present disagreement as a cue for human inspection rather than forcing automatic reconciliation.

Across revisions, traceability is retained by linking each analysis and review to its proposal version. This allows a supervisor to determine whether an identified deficiency disappeared, persisted, or changed in a later draft. It also creates a foundation for future longitudinal evaluation in which measured improvement can be connected to specific feedback and learning resources.

IV. EXPERIMENTAL SETUP AND RESULTS

A. Data and Evaluation Scope

Evaluation focused on the two supervised predictive components for which held-out metrics were available. The weakness dataset contains 2,035 annotated records, with 407 instances used for held-out evaluation. The grading dataset contains 165 scored records, with 33 held-out examples. The semantic retrieval collection contains 2,134 historical feedback records and is used as a search corpus; the current study does not report a labelled retrieval benchmark. Likewise, structural analysis, resource recommendation, and version tracking were validated functionally in the integrated prototype but were not evaluated through a controlled user study. This distinction is important because system implementation evidence should not be interpreted as proof of pedagogical effectiveness.

B. Evaluation Metrics

The weakness classifier is assessed using accuracy and macro-averaged F1. Accuracy measures the proportion of correctly classified held-out instances,

$$\text{Accuracy} = \frac{N_{\text{correct}}}{N_{\text{test}}}. \tag{3}$$

TABLE III
HELD-OUT PREDICTIVE PERFORMANCE

Component	Metric	Result
Grading model	MAE	5.56
Grading model	RMSE	6.50
Grading model	R^2	0.146

Because weakness categories may not be equally represented, accuracy alone can hide poor performance on less frequent classes. Macro-F1 first computes the F1 score for each class and then gives all C classes equal weight,

$$F1_{\text{macro}} = \frac{1}{C}\sum_{c=1}^{C}\frac{2P_cR_c}{P_c+R_c}, \tag{4}$$

where P_c and R_c denote precision and recall for class c. This makes macro-F1 particularly important for interpreting the weakness model.

The proposal-quality regressor is evaluated using MAE, RMSE, and the coefficient of determination. For observed scores y_i and predictions $\hat{y}_i$ over n held-out samples,

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i-\hat{y}_i|, \tag{5}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}. \tag{6}$$

MAE represents average absolute error in score units, while RMSE penalizes larger errors more strongly. The R^2 statistic measures explained variance relative to predicting the mean of the observed scores,

$$R^2 = 1-\frac{\sum_i(y_i-\hat{y}_i)^2}{\sum_i(y_i-\bar{y})^2}. \tag{7}$$

These metrics are reported together because a single aggregate number would not adequately characterize the current model limitations.

C. Weakness Classification Results

The gap between accuracy and macro-F1 indicates uneven performance across weakness categories. In particular, macro-F1 weights categories equally and therefore exposes weak performance on less successfully classified classes even when overall accuracy is higher. The result does not support autonomous diagnosis, but it can still be useful for ranking or surfacing candidate weaknesses for human inspection.

D. Proposal-Grading Results

The TF–IDF Random Forest regressor was evaluated on 33 held-out records. It achieved a Mean Absolute Error (MAE) of 5.56, a Root Mean Squared Error (RMSE) of 6.50, and $R^2 = 0.146$. The low positive R^2 indicates that the model explains only a limited fraction of score variation in unseen examples. The higher RMSE relative to MAE also indicates the presence of larger prediction errors on some records.

These measurements address the predictive-performance question: the current supervised models provide imperfect

predictive signals rather than reliable standalone decisions. Their measured limitations directly motivate the human-in-the-loop policy adopted by the system.

E. Integrated Prototype Validation

The complete prototype was exercised through the end-to-end workflow in Fig. 1. A proposal can be stored as a version, structurally validated, semantically analysed, classified for candidate weaknesses, matched against semantically similar historical feedback, and assigned a baseline quality estimate. The resulting evidence can be assembled into a review draft for supervisor inspection. If revision is requested, a later proposal is stored as a new version without overwriting the previous analysis and review.

This functional validation addresses the engineering feasibility of the proposed review workflow. It shows that heterogeneous evidence can be combined while preserving a clear human decision boundary. It does not, however, establish that the system reduces supervisor workload or improves student learning; those effects require a separate user study.

V. Discussion

The main finding is not that the implemented models can replace a supervisor, but that their limitations can be incorporated into a safer decision-support architecture. The grading model's R^2 of 0.146 is too low to justify autonomous academic decisions. Presenting these outputs as advisory evidence makes the system's use consistent with the actual experimental results.

The architecture also clarifies why a single automated score is insufficient. Structural validation answers whether expected sections appear to be present; semantic concept analysis asks whether important research ideas are actually expressed; the classifier surfaces patterns associated with known weaknesses; retrieval searches for semantically related historical comments; and the regressor provides a coarse numerical signal. None of these outputs alone captures proposal quality, but together they can organize a supervisor's attention.

Semantic retrieval is particularly useful but also potentially misleading. SBERT can identify meaning-level similarity even when wording differs [4]; however, high cosine similarity does not guarantee factual applicability. The current design therefore avoids automatically copying retrieved comments into final feedback. Retrieved items are evidence candidates, and any generated review remains subject to proposal-grounding and supervisor editing.

The mentorship component extends the workflow beyond diagnosis. A student who is told only that a proposal is weak may not know how to improve it. Mapping a specific structural or conceptual deficiency to relevant learning material makes the feedback more actionable and better aligned with formative-assessment principles [1], [2]. Version preservation then allows the system to retain the context of each review cycle instead of reducing supervision to a sequence of disconnected uploads.

From a methodological perspective, the results also support the choice to keep classical models in the current prototype. Prior AES research shows that more complex transformers do not automatically outperform simpler approaches enough to justify their costs in every setting [5]. With only 165 grading records, replacing the regressor with a larger model would not solve the fundamental data limitation. A stronger next step is to expand and curate the dataset, define category-level evaluation clearly, and measure agreement with supervisors before increasing model complexity.

VI. Limitations, Ethics, and Future Work

The current study has four main limitations. First, both supervised datasets are small for the complexity of research-proposal assessment. The weakness classifier exhibits uneven category performance and the grading model explains little score variance. Second, the semantic retrieval module has no labelled relevance benchmark in the current evaluation. Retrieval quality should be measured using supervisor judgements such as precision at k, mean reciprocal rank, or relevance ratings. Third, adaptive resource recommendation and version-based improvement tracking are implemented but not yet validated in a controlled study. Their educational value should therefore be treated as a design hypothesis rather than an empirical result. Fourth, proposal quality is domain-sensitive; a model trained on one institutional context may not generalize to another without recalibration and additional data.

Ethical deployment requires particular care because student proposals may contain personally identifying information, unpublished ideas, and academic records. Institutional implementations should apply access control, secure storage, data minimization, retention policies, and clear disclosure of how student text is processed. Automation bias is another risk: even an explicitly "advisory" score may anchor a supervisor's judgement. Interfaces should therefore expose supporting evidence and model limitations rather than presenting predictions with false precision. Fairness should also be evaluated across relevant student and proposal groups before any automated signal is used at scale, consistent with concerns raised in recent AES research [8].

Future work will expand the weakness and grading datasets, examine class balance, and report per-class precision, recall, and F1 for weakness detection. A labelled retrieval benchmark will be constructed to evaluate semantic feedback matching independently. The most important next evaluation is a supervisor-and-student user study measuring review time, agreement between system findings and expert judgement, usefulness of recommended resources, and improvement across successive proposal versions. More advanced language models may then be compared with the existing lightweight pipeline, but only when they can be evaluated for factual grounding, consistency, fairness, and academic usefulness rather than fluency alone.

VII. Conclusion

This paper presented a human-in-the-loop framework for automated quality assessment and adaptive mentorship of undergraduate research proposals. The system integrates structural validation, semantic concept analysis, TF–IDF/Linear-SVM weakness classification, SentenceTransformer-based feedback retrieval, TF–IDF/Random-Forest proposal-quality regression, evidence-grounded review support, targeted learning resources, and version-aware revision tracking.

The grading model achieved an MAE of 5.56, RMSE of 6.50, and $R^2 = 0.146$ on held-out data. These results are useful precisely because they make the system's limits explicit: the predictive components are not sufficiently reliable to replace academic judgement. Instead, they are used to organize evidence and support a supervisor-controlled review process. The work therefore provides a practical basis for studying responsible AI-assisted research supervision in which automation supports repetitive analysis and adaptive guidance while the human supervisor retains final authority.

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